Advanced Econometrics — MPE (Insper)

Problem set: *L03 — Introduction to Bayesian learning (Classes 3+4): textbook exercises*

Instructions. Every question is a textbook exercise from the "Practice" list of Classes 3+4 in `Map/class-plan.md`, reproduced verbatim from the book PDF with its source line; Portuguese-language books (Bolfarine & Sandoval; Morettin & Singer) are quoted in Portuguese. When an exercise refers to an equation, table or exercise elsewhere in the book, that material is reproduced as *context*. Show every step. Computational items are solved in Python (NumPy/SciPy only, fixed seeds); every number printed in the solutions is reproduced and asserted by `Listas/L03-book-exercises-check.py`. Greenberg's gamma $G(\alpha,\beta)$ and Bolfarine's $Gama(\alpha,\beta)$ both use the rate form $\beta^{\alpha}\theta^{\alpha-1}e^{-\beta\theta}/\Gamma(\alpha)$, mean α/β; $IG(\alpha,\beta)$ is the inverse gamma with density $\beta^{\alpha}y^{-(\alpha+1)}e^{-\beta/y}/\Gamma(\alpha)$ (Greenberg, App. A.1.7–A.1.8).

Block 1 — Greenberg (2007)

(Chs. 2–8 • 15 questions)

Question 1. Greenberg 2.2 — gamma–Poisson conjugacy.
Source: Greenberg, E. (2007), *Introduction to Bayesian Econometrics*, Cambridge University Press, Exercise 2.2, p. 19 — reproduced verbatim.

2.2 The Poisson distribution has probability mass function

$$p(y_i|\theta) = \frac{\theta^{y_i}e^{-\theta}}{y_i!}, \qquad \theta > 0,\ y_i = 0, 1, \ldots,$$

and let $y_1, \ldots, y_n$ be a random sample from this distribution.

(a) Show that the gamma distribution $G(\alpha, \beta)$ is a conjugate prior distribution for the Poisson distribution.

(b) Show that $\bar{y}$ is the MLE for θ.

(c) Write the mean of the posterior distribution as a weighted average of the mean of the prior distribution and the MLE.

(d) What happens to the weight on the prior mean as n becomes large?

Work:

Question 2. Greenberg 2.4 — Pareto–uniform conjugacy.
Source: Greenberg, E. (2007), *Introduction to Bayesian Econometrics*, Cambridge University Press, Exercise 2.4, p. 19 — reproduced verbatim.

2.4 Consider the uniform distribution with density function $f(y_i|\theta) = 1/\theta$, $0 \le y_i \le \theta$, and θ unknown.

(a) Show that the Pareto distribution,

$$\pi(\theta) = \begin{cases} ak^a\theta^{-(a+1)}, & \theta \ge k,\ a > 0, \\ 0, & \text{otherwise}, \end{cases}$$

is a conjugate prior distribution for the uniform distribution.

(b) Show that $\hat{\theta} = \max(y_1, \ldots, y_n)$ is the MLE of θ, where the y_i are a random sample from $f(y_i|\theta)$.

(c) Find the posterior distribution of θ and its expected value.

Work:

Question 3. Greenberg 3.3 — normal predictive distribution.
Source: Greenberg, E. (2007), *Introduction to Bayesian Econometrics*, Cambridge University Press, Exercise 3.3, p. 39 — reproduced verbatim.

> **3.3** Compute the predictive distribution for y_{n+1} if the y_i have independent normal distributions $N(\mu, 1)$, where the prior distribution for μ is $N(\mu_0, \sigma_0^2)$.

Work:

Question 4. Greenberg 3.4 — sequential updating, Poisson model.
Source: Greenberg, E. (2007), *Introduction to Bayesian Econometrics*, Cambridge University Press, Exercise 3.4, p. 39 — reproduced verbatim.

> **3.4** Explicitly verify the updating feature of the posterior distribution for the case where y has a Poisson distribution with parameter θ. Choose $G(\alpha, \beta)$ as the prior distribution, and consider observing a total of s_1 events in the first of two independent experiments and s_2 in the second.

Work:

Question 5. Greenberg 3.2 — Poisson–gamma predictive distribution.
Source: Greenberg, E. (2007), *Introduction to Bayesian Econometrics*, Cambridge University Press, Exercise 3.2, p. 39 — reproduced verbatim.

> **3.2** Compute the predictive distribution for y_{n+1} if the y_i have independent Poisson distributions with parameter θ, given that the first n experiments yielded $y_1, y_2, \ldots, y_n$ events, respectively. Assume the prior distribution $G(\alpha, \beta)$.

Work:

Question 6. Greenberg 3.8 — Bayes factor for two typists.
Source: Greenberg, E. (2007), *Introduction to Bayesian Econometrics*, Cambridge University Press, Exercise 3.8, p. 40 — reproduced verbatim.

> **3.8** Suppose the number of typographical errors per page has a Poisson distribution $P(\theta)$. Sam types the first m pages of a manuscript and makes e_1 errors in total, and Levi types the last m pages and makes e_2 errors. Let Sam's error rate be θ_1 and Levi's error rate be θ_2. To compare model $M_1 : \theta_1 = \theta_2 = \theta$ with model $M_2 : \theta_1 \neq \theta_2$, produce a table like Table 3.2 for $m = 100$ with 10 total errors and for $m = 200$ with 20 total errors and Sam's error proportions of 0.9, 0.8, 0.7, 0.6, and 0.5. Take $G(1, 1)$ as a prior for θ_1 and θ_2.

Context the item refers to (not part of the exercise; Greenberg, Table 3.2, p. 38, reproduced verbatim).

Table 3.2. Bayes Factors for Selected Possible Outcomes

Michaela Proportion heads	Lila Proportion heads	$\log_{10}$(Bayes factor) $m = 10$	$m = 100$
0.1	0.9	−2.506	−30.775
0.2	0.8	−1.200	−15.793
0.3	0.7	−0.348	−6.316
0.4	0.6	0.138	−0.975
0.5	0.5	0.297	0.756

Work:

Question 7. Greenberg 4.5 — marginal likelihood of the conjugate normal regression.
Source: Greenberg, E. (2007), *Introduction to Bayesian Econometrics*, Cambridge University Press, Exercise 4.5, p. 58 — reproduced verbatim.

4.5 Since the posterior distribution of (β, σ^2) in Section 4.3 is of known form, it is possible to compute its marginal likelihood analytically. Start from the definition of the marginal likelihood:

$$\begin{aligned}
&\int f(y|\beta,\sigma^2)\pi(\beta,\sigma^2)\,d\beta\,d\sigma^2 \\
&= \int \left(\frac{1}{2\pi\sigma^2}\right)^{n/2} \exp\left[-\frac{1}{2\sigma^2}(y-X\beta)'(y-X\beta)\right] \left(\frac{1}{2\pi\sigma^2}\right)^{K/2} \\
&\quad \times \frac{1}{|B_0|^{1/2}} \exp\left[-\frac{1}{2\sigma^2}(\beta-\beta_0)'B_0^{-1}(\beta-\beta_0)\right] \\
&\quad \times \frac{(\delta_0/2)^{\alpha_0/2}}{\Gamma(\alpha_0/2)} \left(\frac{1}{\sigma^2}\right)^{\alpha_0/2+1} \exp\left[-\frac{\delta_0}{2\sigma^2}\right] d\beta\,d\sigma^2 \\
&= \left(\frac{1}{2\pi}\right)^{n/2} \frac{(\delta_0/2)^{\alpha_0/2}}{\Gamma(\alpha_0/2)} \frac{1}{|B_0|^{1/2}} \int \left\{ \left(\frac{1}{2\pi\sigma^2}\right)^{K/2} \right. \\
&\quad \left. \times \exp\left[-\frac{1}{2\sigma^2}(\beta-\bar{\beta})'B_1^{-1}(\beta-\bar{\beta})\right] d\beta \right\} \left\{ \left(\frac{1}{\sigma^2}\right)^{\alpha_1/2+1} \exp\left[-\frac{\delta_1}{2\sigma^2}\right] d\sigma^2 \right\},
\end{aligned}$$

where we have used (4.5). Now integrate the first pair of curly braces as a multivariate normal and the second as an inverted gamma to find this expression for the marginal likelihood:

$$\left(\frac{1}{\pi}\right)^{n/2} \frac{|B_1|^{1/2}}{|B_0|^{1/2}} \frac{\Gamma(\alpha_1/2)}{\Gamma(\alpha_0/2)} \frac{\delta_0^{\alpha_0/2}}{\delta_1^{\alpha_1/2}}.$$

Context the item refers to (not part of the exercise; Greenberg §4.3, pp. 45–46, equations (4.2) and (4.6), reproduced verbatim).

$$\begin{aligned}
\pi(\beta,\sigma^2) &= \pi(\beta|\sigma^2)\pi(\sigma^2) \\
&= N_K(\beta|\beta_0,\sigma^2 B_0)\, IG(\sigma^2|\alpha_0/2,\delta_0/2),
\end{aligned} \tag{4.2}$$

[...] where the updated parameters are

$$\begin{aligned}
B_1 &= (X'X + B_0^{-1})^{-1} \\
\bar{\beta} &= B_1(X'y + B_0^{-1}\beta_0) \\
\alpha_1 &= \alpha_0 + n \\
\delta_1 &= \delta_0 + y'y + \beta_0' B_0^{-1}\beta_0 - \bar{\beta}' B_1^{-1}\bar{\beta}.
\end{aligned} \tag{4.6}$$

Work:

Question 8. Greenberg 5.4 — accept–reject sampling of Beta(3, 3).
Source: Greenberg, E. (2007), *Introduction to Bayesian Econometrics*, Cambridge University Press, Exercise 5.4, p. 75 — reproduced verbatim.

> **5.4** Draw 500 observations from Beta(3, 3) using algorithm 5.3. Compute the mean and variance of the sample and compare them to the true values.
>
> *Context the item refers to (not part of the exercise; Greenberg §5.3, p. 67, Algorithm 5.3, reproduced verbatim).*
>
> **Algorithm 5.3: Beta distribution**
> **5.1** Draw u_1 and u_2 from $U(0, 1)$.
> **5.2** If
> $$u_2 \leq \frac{\text{Beta}(3,3)^{-1} u_1^2 (1-u_1)^2}{1.8750},$$
> return $y = u_1$ as a draw from the target distribution. Otherwise reject it and return to 1.

Work:

Question 9. Greenberg 6.2 — the two-state Markov chain.
Source: Greenberg, E. (2007), *Introduction to Bayesian Econometrics*, Cambridge University Press, Exercise 6.2, pp. 87–88 — reproduced verbatim.

> **6.2** Let
> $$P = \begin{pmatrix} 1-\alpha & \alpha \\ \beta & 1-\beta \end{pmatrix}. \tag{6.10}$$
>
> **(a)** Find the invariant distribution if $0 < \alpha, \beta < 1$. Is the process aperiodic? Is it irreducible? What is $\lim P^n$ in this case?
>
> **(b)** Suppose $\alpha = \beta = 0$. Is there a unique invariant distribution? Is the process aperiodic? Is it irreducible? What is $\lim P^n$?
>
> **(c)** Suppose $\alpha = \beta = 1$. Is there a unique invariant distribution? Is the process aperiodic? Is it irreducible? What is $\lim P^n$?

Work:

Question 10. Greenberg 7.1 — full conditionals of the normal model.
Source: Greenberg, E. (2007), *Introduction to Bayesian Econometrics*, Cambridge University Press, Exercise 7.1, p. 105 — reproduced verbatim.

7.1 Verify (7.1) and (7.2).

Context the item refers to (not part of the exercise; Greenberg §7.1.1, p. 93, reproduced verbatim).
Let $y_i \sim N(\mu, h^{-1})$, $i = 1, \ldots, n$, be independently distributed, where the distribution has been parameterized in terms of the precision. We assume the conditionally conjugate priors $\mu \sim N(\mu_0, h_0^{-1})$ and $h \sim G(\alpha_0/2, \delta_0/2)$. [...] From this specification, we have

$$\pi(\mu, h|y) \propto h^{n/2} \exp\left[-\frac{h}{2}\sum(y_i - \mu)^2\right] \exp\left[-\frac{h_0}{2}(\mu - \mu_0)^2\right] \times h^{\alpha_0/2-1} \exp\left[-\frac{\delta_0 h}{2}\right].$$

From here, it is easy to derive the conditional posterior distribution of h,

$$\pi(h|\mu, y) \propto h^{(\alpha_0+n)/2-1} \exp\left[-h\frac{\delta_0 + \sum(y_i - \mu)^2}{2}\right]. \tag{7.1}$$

[...] We complete the square in μ to obtain

$$\pi(\mu|h, y) \propto \exp\left[-\frac{h_0 + hn}{2}\left(\mu - \frac{h_0\mu_0 + hn\bar{y}}{h_0 + hn}\right)^2\right], \tag{7.2}$$

which should be recognized as $N[(h_0\mu_0 + hn\bar{y})/(h_0 + hn), (h_0 + hn)^{-1}]$ [...].

Work:

Question 11. Greenberg 7.3 — Gibbs sampler for the coal-mining switch point.
Source: Greenberg, E. (2007), *Introduction to Bayesian Econometrics*, Cambridge University Press, Exercise 7.3, p. 106 — reproduced verbatim.

7.3 Construct a Gibbs algorithm to analyze the Poisson model with unknown switch point. Given the specification in Equations (7.3) and (7.4), show that

$$\pi(\theta_1, \theta_2, k|y) \propto \theta_1^{\alpha_{10}-1} e^{-\beta_{10}\theta_1} \theta_2^{\alpha_{20}-1} e^{-\beta_{20}\theta_2} \prod_{i=1}^{k} e^{-\theta_1} \theta_1^{y_i} \prod_{i=k+1}^{n} e^{-\theta_2} \theta_2^{y_i},$$

and verify that

$$\theta_1|y, k \sim G\Big(\alpha_{10} + \sum_{1}^{k} y_i,\ \beta_{10} + k\Big),$$

$$\theta_2|k, y \sim G\Big(\alpha_{20} + \sum_{k+1}^{n} y_i,\ \beta_{20} + n - k\Big),$$

$$\pi(k|y, \theta_1, \theta_2) = \frac{e^{k(\theta_2-\theta_1)}(\theta_1/\theta_2)^{\sum_1^k y_i}}{\sum_{k=1}^{n} e^{k(\theta_2-\theta_1)}(\theta_1/\theta_2)^{\sum_1^k y_i}}, \qquad k = 1, \ldots, n.$$

Apply your algorithm to the mining disaster data analyzed by Carlin et al. (1992a); the data may be found in the article. Your analysis should generate marginal posterior distributions for θ_1, θ_2, $p(k)$, and the marginal likelihood computed by the method of Chib (1995). For simplicity, set $\alpha_{10} = \alpha_{20} = 0.5$ and $\beta_{10} = \beta_{20} = 1.0$. (Curiously, although these data have been analyzed from the Bayesian viewpoint in several papers, none of them present historical material that might shed light on when changes in parameters took place. For example, changes in government regulation, technology, or the location of mines might be expected to affect the probability of a disaster. A more general version of the model permits multiple switch points if necessary.)

Context the item refers to (not part of the exercise; Greenberg §7.1.1, pp. 93–94, equations (7.3)–(7.4), reproduced verbatim).

$$p(y_i) = \begin{cases} \dfrac{e^{-\theta_1}\theta_1^{y_i}}{y_i!}, & \text{for } i = 1, \ldots, k, \\ \dfrac{e^{-\theta_2}\theta_2^{y_i}}{y_i!}, & \text{for } i = k+1, \ldots, n, \end{cases} \tag{7.3}$$

where $y_i = 0, 1, \ldots$, and the switch point k is unknown. The specification is completed by assigning the conditionally conjugate priors,

$$\theta_1 \sim G(\alpha_{10}, \beta_{10}), \quad \theta_2 \sim G(\alpha_{20}, \beta_{20}), \quad \pi(k = j) = 1/n, \; j = 1, \ldots, n. \tag{7.4}$$

Work:

Question 12. Greenberg 7.5 — random-walk Metropolis–Hastings for the Laplace distribution.
Source: Greenberg, E. (2007), *Introduction to Bayesian Econometrics*, Cambridge University Press, Exercise 7.5, p. 106 — reproduced verbatim.

7.5 Construct a random walk MH sampler to generate a sample of 10,000 from the Laplace distribution, $f(x) = (1/2)\, e^{-|x|}$, $-\infty < x < \infty$. Use $u \sim N(0, \sigma^2)$ to generate proposals $y = x^{(g-1)} + u$. Compare the mean, variances, and autocorrelations of the generated series for various values of σ^2.

Work:

Question 13. Greenberg 7.8 — MH within Gibbs with a tailored truncated-t proposal.
Source: Greenberg, E. (2007), *Introduction to Bayesian Econometrics*, Cambridge University Press, Exercise 7.8, p. 107 — reproduced verbatim.

> **7.8** Generalize Exercise 7.7 to allow for an unknown variance, $u_i \sim N(0, \sigma^2)$, with prior distribution $\sigma^2 \sim \text{IG}(5/2, 3/2)$. Verify that $\sigma^2|y, \beta$ has an inverse gamma distribution, but that the distribution of $\beta|y, \sigma^2$ is not standard. Devise a two-block algorithm in which $\sigma^2|y, \beta$ is generated in a Gibbs step and β is generated by a tailored kernel where $\hat{\beta}$ depends on σ^2, so that the mean and scale factor of the truncated Student-t distribution must be recomputed at each step. Choose a value of σ^2 from its prior distribution, and generate and analyze a data set as in Exercise 7.7.
>
> *Context the item refers to (not part of the exercise; Greenberg, Exercise 7.7, pp. 106–107, reproduced verbatim).*
>
> **7.7** Consider the model
>
> $$y_i = \beta x_i + u_i, \quad u_i \sim N(0, 1), \quad i = 1, \ldots, n,$$
>
> with the gamma prior distribution $\beta \sim G(2, 1)$, $\beta > 0$. Verify the posterior distribution
>
> $$\pi(\beta|y) \propto \beta \exp[-\beta] \exp\left[-\frac{1}{2}\sum_{i=1}^{n}(y_i - \beta x_i)^2\right] 1(\beta > 0).$$
>
> Note that this distribution does not have a standard form. Construct an MH algorithm to sample from this distribution with an independence kernel, where the kernel is a Student-t distribution truncated to the region $(0, \infty)$, with five degrees of freedom, mean equal to the value of β that maximizes the posterior distribution ($\hat{\beta}$), and scale factor equal to the negative inverse of the second derivative of the log posterior distribution evaluated at $\hat{\beta}$. Verify that
>
> $$\hat{\beta} = \frac{\left(\sum x_i y_i - 1\right) + \sqrt{\left(\sum x_i y_i - 1\right)^2 + 4\sum x_i^2}}{2\sum x_i^2}$$
>
> and that the scale factor is $(1/\hat{\beta}^2 + \sum x_i^2)^{-1}$. Generate a data set by choosing $n = 50$, x_i from $N(0, 1)$, and a value of β from its prior distribution. Write a program to implement your algorithm and see how well β is determined. You may try larger values of n to explore the effect of sample size, and, depending on the acceptance rate, you may wish to change the scale factor.

Work:

Question 14. Greenberg 8.1 — Bayesian regression for birth weight.
Source: Greenberg, E. (2007), *Introduction to Bayesian Econometrics*, Cambridge University Press, Exercise 8.1, p. 132 — reproduced verbatim.

> **8.1** Download the data in "Birth weight II" from the StatLabs Web site, www.stat.berkeley.edu/users/statlabs/labs.html. The variables are birth weight (ounces, 999 unknown), gestation (length of pregnancy in days, 999 unknown), parity (0 = firstborn, 9 is unknown), age (mother's age in years), height (mother's height in inches), weight (mother's prepregnancy weight in pounds), smoke (0 not now, 1 yes now, 9 = unknown).
>
> Specify a linear regression, where the response variable is birth weight, and the covariates are a constant and the remaining variables in the data set. Eliminate observations for which you do not have complete information on the variables. Specify prior distributions for the coefficients and the variance (or precision). Summarize the results for the dummy variable for smoking. Does smoking by the mother appear to affect the baby's birth weight?

Work:

Question 15. Greenberg 8.8 — logit posterior and latent data.
Source: Greenberg, E. (2007), *Introduction to Bayesian Econometrics*, Cambridge University Press, Exercise 8.8, pp. 132–133 — reproduced verbatim.

> **8.8** Verify that the posterior distribution of the logit model with a normal prior (Section 8.2.3) has the form
>
> $$\pi(\beta|y) \propto \prod \left(\frac{\exp[x_i'\beta]}{1+\exp[x_i'\beta]}\right)^{y_i} \left(\frac{1}{1+\exp[x_i'\beta]}\right)^{1-y_i} \times \exp\left[-\frac{1}{2}(\beta-\beta_0)'B_0^{-1}(\beta-\beta_0)\right],$$
>
> and show that including the latent data in the sampler does not simplify the posterior distribution.
>
> *Context the item refers to (not part of the exercise; Greenberg §8.2.3, pp. 126–127, reproduced verbatim).*
>
> The binary logit model is another model for binary data. It is based on
>
> $$P(y_i = 1) = P_i = \frac{\exp[x_i'\beta]}{1+\exp[x_i'\beta]},$$
>
> [...] This model can be written in the form of latent data as
>
> $$y_i^* = x_i'\beta + u_i,$$
>
> where u_i has the *logistic* distribution, $P(U_i \le u_i) = \Lambda(u_i) = e^{u_i}/(1+e^{u_i})$ and density function $\lambda(u_i) = e^{u_i}/(1+e^{u_i})^2$.

Work:

Block 2 — Tsay (2010)

(Ch. 12 • 3 questions)

Question 16. Tsay 12.1 — normal mean, one observation.
Source: Tsay, R. S. (2010), *Analysis of Financial Time Series*, 3rd ed., Wiley, Exercise 12.1, p. 670 — reproduced verbatim.

> **12.1.** Suppose that x is normally distributed with mean μ and variance 4. Assume that the prior distribution of μ is also normal with mean 0 and variance 25. What is the posterior distribution of μ given the data point x?

Work:

Question 17. Tsay 12.2 — Gibbs conditionals for regression with AR(p) errors.
Source: Tsay, R. S. (2010), *Analysis of Financial Time Series*, 3rd ed., Wiley, Exercise 12.2, p. 670 — reproduced verbatim.

> **12.2.** Consider the linear regression model with time series errors in Section 12.5. Assume that z_t is an AR(p) process (i.e., $z_t = \phi_1 z_{t-1} + \cdots + \phi_p z_{t-p} + a_t$). Let $\boldsymbol{\phi} = (\phi_1, \ldots, \phi_p)'$ be the vector of AR parameters. Derive the conditional posterior distributions of $f(\boldsymbol{\beta}|\boldsymbol{Y}, \boldsymbol{X}, \boldsymbol{\phi}, \sigma^2)$, $f(\boldsymbol{\phi}|\boldsymbol{Y}, \boldsymbol{X}, \boldsymbol{\beta}, \sigma^2)$, and $f(\sigma^2|\boldsymbol{Y}, \boldsymbol{X}, \boldsymbol{\beta}, \boldsymbol{\phi})$ using the conjugate prior distributions, that is, the priors are
>
> $$\boldsymbol{\beta} \sim N(\boldsymbol{\beta}_o, \boldsymbol{\Sigma}_o), \qquad \boldsymbol{\phi} \sim N(\boldsymbol{\phi}_o, \boldsymbol{A}_o), \qquad (v\lambda)/\sigma^2 \sim \chi^2_v.$$
>
> *Context the item refers to (not part of the exercise; Tsay §12.5, p. 624, equation (12.6), reproduced verbatim).*
>
> The model becomes
>
> $$y_t = \boldsymbol{x}_t'\boldsymbol{\beta} + z_t, \qquad z_t = \phi z_{t-1} + a_t, \qquad t = 1, \ldots, n, \tag{12.6}$$
>
> where n is the sample size. [...] $\{a_t\}$ being a sequence of independent and identically distributed normal random variables with mean zero and variance σ^2.

Work:

Question 18. Tsay 12.3 — two consecutive missing values in an AR(p).
Source: Tsay, R. S. (2010), *Analysis of Financial Time Series*, 3rd ed., Wiley, Exercise 12.3, p. 670 — reproduced verbatim.

> **12.3.** Consider the linear AR(p) model in Section 12.6.1. Suppose that x_h and x_{h+1} are two missing values with a joint prior distribution being multivariate normal with mean $\boldsymbol{\mu}_o$ and covariance matrix $\boldsymbol{\Sigma}_o$. Other prior distributions are the same as that in the text. What is the conditional posterior distribution of the two missing values?
>
> *Context the item refers to (not part of the exercise; Tsay §12.6.1, pp. 629–630, reproduced verbatim).*
>
> For ease in presentation, consider an AR(p) time series
>
> $$x_t = \phi_1 x_{t-1} + \cdots + \phi_p x_{t-p} + a_t, \tag{12.15}$$
>
> where $\{a_t\}$ is a Gaussian white noise series with mean zero and variance σ^2. [...] If we assume that the prior distributions are
>
> $$\boldsymbol{\phi} \sim N(\boldsymbol{\phi}_o, \boldsymbol{\Sigma}_o), \qquad x_h \sim N(\mu_o, \sigma_o^2), \qquad \frac{v\lambda}{\sigma^2} \sim \chi^2_v,$$
>
> where the hyperparameters are known [...].

Work:

Block 3 — Bolfarine & Sandoval (2010) *(Chs. 4–5 • 5 questions)*

Question 19. Bolfarine & Sandoval 4.3 — risk and Bayes risk, binomial.
Source: Bolfarine, H. and Sandoval, M. C. (2010), *Introdução à Inferência Estatística*, 2nd ed., SBM, Exercício 4.3, p. 71 — reproduced verbatim (Portuguese original).

> **4.3.** Considere uma única observação da variável aleatória $X \sim Binomial(m, \theta)$. Seja $l(\theta, d) = (\theta - d)^2$.
> (i) Encontre o risco de $d(X) = X/m$.
> (ii) Encontre o risco de Bayes de $d(X)$ em (i), com relação a priori $\pi(\theta) = 1, 0 \leq \theta \leq 1$.

Work:

Question 20. Bolfarine & Sandoval 4.5 — Bayes risk of X under a gamma prior.
Source: Bolfarine, H. and Sandoval, M. C. (2010), *Introdução à Inferência Estatística*, 2nd ed., SBM, Exercício 4.5, p. 71 — reproduced verbatim (Portuguese original).

> **4.5.** Seja uma única observação da distribuição $Poisson(\theta)$. Encontre o risco de Bayes do estimador $d(X) = X$, com relação à perda quadrática e a priori $Gama(\alpha, \beta)$.

Work:

Question 21. Bolfarine & Sandoval 4.9 — Bayes estimator of $P_\theta(X = 0)$.
Source: Bolfarine, H. and Sandoval, M. C. (2010), *Introdução à Inferência Estatística*, 2nd ed., SBM, Exercício 4.9, p. 72 — reproduced verbatim (Portuguese original).

> **4.9.** Em uma área de reflorestamento, o número de árvores de determinada espécie, por hectare, com certa doença tem uma distribuição $Poisson(\theta)$. A distribuição a priori de θ é exponencial com média igual a 1. Encontre o estimador de Bayes de $P_\theta(X = 0)$ com relação à perda quadrática..

Work:

Question 22. Bolfarine & Sandoval 4.10 — Pareto prior for $U(0,\theta)$.
Source: Bolfarine, H. and Sandoval, M. C. (2010), *Introdução à Inferência Estatística*, 2nd ed., SBM, Exercício 4.10, p. 72 — reproduced verbatim (Portuguese original).

> **4.10.** Sejam $X_1,\ldots,X_n$ uma amostra aleatória da distribuição $U(0,\theta)$. Suponhamos que θ seja uma variável aleatória com função de densidade de probabilidade (Pareto)
>
> $$\pi(\theta) = \begin{cases} ba^b/\theta^{b+1}, & \theta \geq a, \\ 0, & \theta < a, \end{cases}$$
>
> Encontre a distribuição a posteriori de θ e o estimador de Bayes de θ com relação à perda quadrática.

Work:

Question 23. Bolfarine & Sandoval 5.11 — equal-tailed Bayesian interval for a Poisson mean.
Source: Bolfarine, H. and Sandoval, M. C. (2010), *Introdução à Inferência Estatística*, 2nd ed., SBM, Exercício 5.11, p. 89 — reproduced verbatim (Portuguese original).

> **5.11.** Sejam $X_1,\ldots,X_n$ uma amostra aleatória de tamanho n da distribuição $Poisson(\theta)$, com priori
>
> $$\pi(\theta) = e^{-\theta}, \quad \theta > 0.$$
>
> Construa um intervalo de confiança Bayesiano simétrico para θ com $\gamma = 0{,}95$. Se $n = 10$ e $\sum_{i=1}^{n} X_i = 18$, como fica o intervalo?

Work:

Block 4 — Morettin & Singer (2021) *(Apps. B–C • 3 questions)*

Question 24. Morettin & Singer B.16 — accept–reject for a truncated gamma.
Source: Morettin, P. A. and Singer, J. M. (2021), *Estatística e Ciência de Dados* (preliminary version, April 2021), Apêndice B, Exercício 16, p. 478 — reproduced verbatim (Portuguese original).

> 16) Considere a distribuição gama truncada $\pi(x) \propto x^2 \exp(-x)$, $0 \leq x \leq 1$. Simule uma amostra de tamanho 1.000 dessa distribuição usando o método da aceitação-rejeição com $g(x) = \exp(-x)$.

Work:

Question 25. Morettin & Singer B.19 — accept–reject for discrete variables.
Source: Morettin, P. A. and Singer, J. M. (2021), *Estatística e Ciência de Dados* (preliminary version, April 2021), Apêndice B, Exercício 19, p. 478 — reproduced verbatim (Portuguese original).

> 19) Considere o método da aceitação-rejeição no caso de variáveis discretas. Especificamente, suponha que queiramos simular valores de uma variável aleatória X, com probabilidades $\pi_j = P(X = j)$ e dispomos de um método para simular valores da variável aleatória Y, com probabilidades $g_j = P(Y = j)$. Suponha ainda, que exista uma constante c, tal que $\pi_j \leq cg_j$, para todo j. Usando o algoritmo de aceitação-rejeição, mostre que $P(\text{aceitar}) = 1/c$ e que o número de iterações necessárias para obter X é uma variável aleatória com distribuição geométrica com média c. Com esse resultado, mostre que $P(X = j) = \pi_j$.

Work:

Question 26. Morettin & Singer C.5 — chained data (data augmentation) algorithm.
Source: Morettin, P. A. and Singer, J. M. (2021), *Estatística e Ciência de Dados* (preliminary version, April 2021), Apêndice C, Exercício 5, p. 489 — reproduced verbatim (Portuguese original).

> 5) **Algoritmo de Dados Encadeados** (*Chain Data Algorithm*)
> Numa abordagem um pouco diferente do algoritmos para dados aumentados, iteramos sucessivamente de $p(\mathbf{z}|\boldsymbol{\theta}, \mathbf{y})$ e $p(\boldsymbol{\theta}|\mathbf{y}, \mathbf{z})$ para obter uma sequência de realizações $(\mathbf{z}^{(1)}, \boldsymbol{\theta}^{(1)}), \ldots, (\mathbf{z}^{(n)}, \boldsymbol{\theta}^{(n)})$ com as simulações de uma variável realizadas condicionalmente aos valores atuais da outra variável. Esperamos que a distribuição de equilíbrio dessa cadeia seja a distribuição conjunta $p(\mathbf{z}, \boldsymbol{\theta}|\mathbf{y})$. Este algoritmo é um caso especial do amostrador de Gibbs, descrito na Nota de Capítulo 2 do Apêndice B. Normalmente, estaremos interessados nas distribuições marginais.

Para os dados do Exemplo C.2, obtenha as sequências de realizações de Z e θ. Simule $n = 200$ valores, desprezando os primeiros 50 valores, conhecidos como **valores de aquecimento**) (*burn-in*).

Context the item refers to (not part of the exercise; Morettin & Singer, Exemplo C.2, p. 480, excerpt, reproduced verbatim).

Exemplo C.2. Retomemos o problema da ligação genética discutido no Exemplo B.16, segundo o qual os dados $\mathbf{y} = (y_1, y_2, y_3, y_4)^\top = (125, 18, 20, 34)^\top$ ocorrem com probabilidades $\pi_1 = (2+\theta)/4$, $\pi_2 = \pi_3 = (1-\theta)/4$, $\pi_4 = \theta/4$, $0 < \theta < 1$. [...] Admitamos que $\mathbf{x} = (x_1, \ldots, x_5)^\top$ sejam os dados completos que ocorrem, respectivamente, com probabilidades $(1/2, \theta/4, (1-\theta)/4, (1-\theta)/4, \theta/4)$, mas que só observamos $y_1 = x_1 + x_2$, $y_2 = x_3$, $y_3 = x_4$ e $y_4 = x_5$, [...] Podemos simplificar as expressões acima obtendo

$$L(\theta|\mathbf{y}) \propto (2+\theta)^{y_1}(1-\theta)^{y_2+y_3}\theta^{y_4}, \tag{C.2}$$

e

$$L(\theta|\mathbf{X}) \propto \theta^{x_2+x_5}(1-\theta)^{x_3+x_4}. \tag{C.3}$$

Work: